

Tanish Dalal

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EDUCATION

University of Michigan

B.S.E. in Computer Engineering

GPA: 4.00/4.00 Honors: University Honors; Dean's List

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Architecture, Discrete Mathematics, Linear Algebra, Logic Design

Ann Arbor, MI

Aug. 2024 – May 2028

EXPERIENCE

Software Engineering Intern

ADNOC Distribution, EVCI & H2 Department

May 2025 – Jul. 2025

Abu Dhabi, UAE

- Developed a React.js interface for EV-charger session, consumption, and revenue metrics, reducing troubleshooting time by 25% for 30+ engineers; integrated and validated APIs with automated JavaScript tests.
- Resolved a long-standing charger uptime issue by isolating overheating as the root cause; preventing \$15k in lost revenue.
- Built a Python ARIMA prototype model predicting temperature impact on charger uptime with 8% MAPE.
- Authored a 25-page EV-charger installation guide adopted by 35+ staff across 5 departments; evaluated vendors (Sinexcel, Teison) and proposed deployment strategies.

Robotics Engineering Intern

ABB

Jun. 2023 – Jul. 2023

Abu Dhabi, UAE

- Developed and tested welding robot paths in ABB RobotStudio, optimizing RAPID routines for approach, weld, and retract motions.
- Designed collision-detection and waypoint-refinement logic to maintain TCP clearance; verified reliability in 50+ RobotStudio simulations with 0 collisions.
- Automated collision checks around jigs and fixtures in RobotStudio, cutting path rework by 30%.

PROJECTS

Computer Vision Lead & Co-Founder — Michigan Robomasters

C++, ROS 2, RViz

Aug. 2025 – Present

- Developed embedded motor control firmware in C/C++ for STM32 microcontrollers, implementing M3508 drive motors and GM6020 gimbal controls using CAN bus and UART protocols.
- Programmed directional movement logic and precision gimbal targeting for coordinated multi-axis chassis control.
- Implemented PID control algorithms for motor velocity and position regulation, achieving [X]
- Leading CV subteam of [Z] members in developing real-time armor plate detection and auto-aiming systems using Python and C++.
- Designing YOLO/OpenCV-based object detection pipeline for enemy robot tracking, targeting [X] FPS processing rate with [Y]
- Implementing Kalman filtering algorithms for target prediction and lead compensation in autonomous shooting systems.

Arm IK Controls Developer — Michigan Mars Rover

C++, ROS 2, RViz

Aug. 2024 – Present

- Implemented inverse kinematics (IK) and keyboard teleop for a 5-DOF robotic arm, enabling precise velocity control.
- Developed Cartesian velocity control using a 'carrot-on-a-stick' method for 15% smoother end-effector motion.
- Integrated real-time ROS 2 publishers/subscribers (30 Hz) for joint states, velocity commands, and error handling.
- Enhanced precision and reduced navigation errors by 20% via improved velocity control.
- Onboarded 25 new members on project setup, Ubuntu installation, and simulator configuration.
- Developing a Click IK interface for fast, on-screen end-effector position control.

Algorithm Practice Platform

React, FastAPI, Python

Sep. 2025 – Present

- Designing a full-stack web application using Python (FastAPI) and JavaScript (React) to evaluate algorithm solutions against curated tests, reporting pass/fail status, runtime, and memory usage.
- Developing a FastAPI backend that securely executes Python code with resource limits; currently supports 3 classic problems (arrays, graphs, sorting) and 20+ automated test cases.
- Building a React front end with a live editor and results dashboard; early tests show 150 ms average evaluation time on standard inputs.

TECHNICAL SKILLS

Languages: C/C++ (OOP), Python, Java, JavaScript, Assembly, MATLAB, HTML5/CSS3

Frameworks: React, FastAPI, ROS 2

Libraries: pandas, NumPy, Matplotlib, SciPy

Developer Tools: Git, GitHub Actions (CI/CD), Google Cloud Platform, LaTeX

Testing & Simulation: ModelSim, Pytest, RViz, ABB RobotStudio

Design/CAD: Autodesk Fusion 360, AutoCAD

Hardware/Embedded: Arduino, Verilog, Raspberry Pi